# Ubuntu_Jar_File_Creation_Steps (1) (1).docx

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## Summary

This tutorial provides a comprehensive guide for developing and executing a **WordCount application** within a **Hadoop MapReduce framework** using the Eclipse IDE. The process begins by configuring a Java project with specific **external Hadoop libraries** and defining the three core components: a Mapper to tokenize text, a Reducer to aggregate word frequencies, and a Driver to manage the job's execution. After exporting the code as a **runnable JAR file**, the instructions detail how to prepare the **distributed file system (HDFS)** by creating directories and uploading source data. Finally, the guide covers the essential commands to **initialize Hadoop services** and launch the job to generate a calculated frequency analysis of the input text.



**Keywords:** JAR file creation, Eclipse project setup, Hadoop MapReduce, Java WordCount program, HDFS command execution


## Content

Jar File Creation Steps:1. First Open Eclipse -> then select File->New->Java Project->Name it WordCount-> then Finish. 2. You have to include Reference Libraries for that:Right Click on Project-> then select Build Path-> Click on Configure Build Path In the above figure, you can see the Add External JARs option on the Right Hand Side. Click on it and add the below mention files1. /home/bmsce/hadoop-2.6.0/share/hadoop/mapreduce/2. /home/bmsce/hadoop-2.6.0/share/common/3. Select JRE System Library- goto edit- select JavaSE-1.8 and click apply 4.Create Three Java Classes into the project. Name them WCDriver(having the main function), WCMapper, WCReducer.// Importing librariesimport java.io.IOException;import org.apache.hadoop.io.IntWritable;import org.apache.hadoop.io.LongWritable;import org.apache.hadoop.io.Text;import org.apache.hadoop.mapred.MapReduceBase;import org.apache.hadoop.mapred.Mapper;import org.apache.hadoop.mapred.OutputCollector;import org.apache.hadoop.mapred.Reporter;public class WCMapper extends MapReduceBase implements Mapper<LongWritable,Text, Text, IntWritable> { // Map function public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter rep) throws IOException { String line = value.toString(); // Splitting the line on spaces for (String word : line.split(” “)) { if (word.length() > 0) { output.collect(new Text(word), new IntWritable(1)); } } }} Reducer Code: You have to copy paste this program into the WCReducer Java Class file. // Importing librariesimport java.io.IOException;import java.util.Iterator;import org.apache.hadoop.io.IntWritable;import org.apache.hadoop.io.Text;import org.apache.hadoop.mapred.MapReduceBase;import org.apache.hadoop.mapred.OutputCollector;import org.apache.hadoop.mapred.Reducer;import org.apache.hadoop.mapred.Reporter;public class WCReducer extends MapReduceBase implements Reducer<Text,IntWritable, Text, IntWritable> { // Reduce function public void reduce(Text key, Iterator<IntWritable> value, OutputCollector<Text, IntWritable> output, Reporter rep) throws IOException { int count = 0; // Counting the frequency of each words while (value.hasNext()) { IntWritable i = value.next(); count += i.get(); } output.collect(key, new IntWritable(count)); }}Driver Code: You have to copy paste this program into the WCDriver Java Class file.// Importing librariesimport java.io.IOException;import org.apache.hadoop.conf.Configured;import org.apache.hadoop.fs.Path;import org.apache.hadoop.io.IntWritable;import org.apache.hadoop.io.Text;import org.apache.hadoop.mapred.FileInputFormat;import org.apache.hadoop.mapred.FileOutputFormat;import org.apache.hadoop.mapred.JobClient;import org.apache.hadoop.mapred.JobConf;import org.apache.hadoop.util.Tool;import org.apache.hadoop.util.ToolRunner;public class WCDriver extends Configured implements Tool { public int run(String args[]) throws IOException { if (args.length < 2) { System.out.println(”Please give valid inputs”); return -1; } JobConf conf = new JobConf(WCDriver.class); FileInputFormat.setInputPaths(conf, new Path(args[0])); FileOutputFormat.setOutputPath(conf, new Path(args[1])); conf.setMapperClass(WCMapper.class); conf.setReducerClass(WCReducer.class); conf.setMapOutputKeyClass(Text.class); conf.setMapOutputValueClass(IntWritable.class); conf.setOutputKeyClass(Text.class); conf.setOutputValueClass(IntWritable.class); JobClient.runJob(conf); return 0; } // Main Method public static void main(String args[]) throws Exception { int exitCode = ToolRunner.run(new WCDriver(), args); System.out.println(exitCode); }}Now you have to make a jar file. Right Click on Project-> Click on Export-> Select export destination as Jar File-> Name the jar File(WordCount.jar) -> Click on next -> at last Click on Finish.To run Update the following changes in mapred-site.xml (/home/hadoop/hadoop/etc)<property> <name>yarn.app.mapreduce.am.env</name> <value>HADOOP_MAPRED_HOME=$HADOOP_HOME</value></property><property> <name>mapreduce.map.env</name> <value>HADOOP_MAPRED_HOME=$HADOOP_HOME</value></property><property> <name>mapreduce.reduce.env</name> <value>HADOOP_MAPRED_HOME=$HADOOP_HOME</value></property>To Run MapReduce Program1. hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ start-all.sh (OR)hduser@bmsce-Precision-T1700:/$ su hduserPassword:hduser@bmsce-Precision-T1700:/$ cd /hduser@bmsce-Precision-T1700:/$ cd /usr/local/hadoop/sbinhduser@bmsce-Precision-T1700:/usr/local/hadoop/sbin$ start-all.sh 2. hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC$ jps (OR)hduser@bmsce-Precision-T1700:/usr/local/hadoop/sbin$ jps 3. create a file on Desktop(sample.txt) and type the below lines:hi how are youhow is your jobhow is your familyhow is your brotherhow is your sistersave your file5. View the directory contenthadoop fs -ls /6. Create a directory using the following command. If any directory existing, use the same directory for the command hadoop fs -mkdir /rgs7. Copy the file into HDFS hadoop fs -copyFromLocal D:/sample.txt /rgs/test.txt8.Run the Map Reduce Programhadoop jar /home/hduser/Desktop/Jwordcount.jar WCDriver input output9. hadoop fs -ls /output/10. hadoop fs -cat /output/part-00000